Congruences for the scaled inverse series A393856

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Abstract

For the scaled inverse series A393856 we prove that every positive-index coefficient is one modulo 5. We also prove that a coefficient is odd exactly when its index is a power of two. The source equation, coefficientwise interpretation of its quotient, existence, uniqueness, and the all-index proof are given explicitly.

1 Source equation and theorem

For a positive integer q, consider the normalized series G_q satisfying

$$G_q\left(x - \frac{xG_q(qx)}{q}\right) = x, \qquad G_q = x + O(x^2). \tag{1}$$

The source A393856 of Paul D. Hanna is the case $q = 4$ [1]. Put $a_n = [x^n]G_4$. Its source indices start at one and $a_0 = 0$. We prove

$$a_n \equiv 1 \pmod 5 \qquad (n \geq 1).$$

For every $n \geq 0$ we also prove

$$a_n \text{ is odd} \quad \Longleftrightarrow \quad n = 2^j \text{ for some } j \geq 0.$$

Thus the parity statement includes $n = 0$ and the initial term $n = 1$.

All series are formal: equality and congruence are coefficientwise, and composition $U(V) = \sum_{j\geq 0} u_j V^j$ is used only when $V(0) = 0$. Every coefficient then receives finitely many contributions. A series with constant coefficient one is a multiplicative unit: if $W(0) = 0$, its inverse is $\sum_{j\geq 0}(-W)^j$. This justifies the rational comparison series below without analytic convergence or division in a residue ring.

1.1 Removing the apparent division

If $U(x) = \sum_{n\geq 1} u_n x^n$, define

$$L_q(U)(x) = \sum_{n\geq 1} q^{n-1} u_n x^n.$$

For a nonzero integer q, this is exactly $U(qx)/q$. The coefficient definition makes sense over every commutative ring, without requiring that q be invertible. Equation (1) therefore means

$$G_q(x - xL_q(G_q)(x)) = x. \tag{2}$$

Lemma 1.1. *For any commutative ring R with identity and any $q \in R$, equation (2) has a unique normalized solution in $R[[x]]$. Its construction commutes with ring homomorphisms.*

Proof. Write $G = x + \sum_{n\geq 2} g_n x^n$ and $V = x - xL_q(G)$. The series V is normalized. The coefficient g_n inside V first occurs in degree $n+1$, because of the initial factor x. Thus the coefficient of degree n in V depends only on $g_2, \ldots, g_{n-1}$.

In $G(V) = V + \sum_{j\geq 2} g_j V^j$, the term $g_n V^n$ contributes exactly g_n in degree n. The terms with $j < n$ depend only on earlier coefficients, and those with $j > n$ contribute nothing in that degree. Therefore the equation in degree n has the form $g_n + P_n = 0$, with P_n determined by earlier coefficients and q. This gives existence and uniqueness by recursion, without division. It also shows that applying a ring homomorphism to the coefficients preserves the equation; uniqueness identifies the resulting solution. □

1.2 The congruence and parity theorems

Theorem 1.2. *For every positive integer q,*

$$G_q(x) \equiv \frac{x}{1-x} \pmod{q+1}.$$

In particular, every coefficient of positive degree is congruent to one modulo $q+1$.

Proof. Work in $R = \mathbb{Z}/(q+1)\mathbb{Z}$, where $q = -1$. For $T = x/(1-x)$,

$$L_{-1}(T) = x - x^2 + x^3 - \cdots = \frac{x}{1+x}.$$

Hence

$$x - xL_{-1}(T) = x - \frac{x^2}{1+x} = \frac{x}{1+x},$$

and direct substitution gives

$$T\left(\frac{x}{1+x}\right) = \frac{x/(1+x)}{1-x/(1+x)} = x.$$

Thus T is the normalized solution of the reduced equation. Lemma 1.1 identifies it with the reduction of G_q. This argument also covers composite moduli such as six and eight. □

Theorem 1.3. *If q is a positive even integer, then*

$$G_q(x) \equiv \sum_{j\geq 0} x^{2^j} \pmod 2.$$

Equivalently, its coefficient of degree $n \geq 0$ is odd if and only if n is a power of two. If q is odd, all positive-index coefficients of G_q are odd.

Proof. For even q, reduction modulo two gives $L_q(U) = x$ for every normalized U: all factors q^{n-1} with $n \geq 2$ vanish. Therefore the reduced equation is

$$G_q(x + x^2) = x.$$

Let $S(x) = \sum_{j\geq 0} x^{2^j}$ over $\mathbb{Z}/2\mathbb{Z}$. The strictly increasing exponents make this a well-defined formal series. Repeated squaring in characteristic two gives $(x+x^2)^{2^j} = x^{2^j} + x^{2^{j+1}}$, and hence

$$S(x + x^2) = \sum_{j\geq 0} (x^{2^j} + x^{2^{j+1}}) = x.$$

Every term beyond the first occurs twice and cancels. By uniqueness, the reduction of G_q is S. Its support is exactly the powers of two, including $1 = 2^0$ and excluding zero.

If q is odd, then $q+1$ is even, so Theorem 1.2 already gives oddness of every positive-index coefficient. □

2 Application to A393856

Take $q = 4$ in Theorem 1.2. Its modulus is $q + 1 = 5$, so every positive-index coefficient has the asserted residue. The parameter is even, so Theorem 1.3 gives the precise power-of-two support modulo two. These are both source assertions, with no upper index bound.

3 Formal proof and scope

The accompanying entry file is `lean/Entries/A393856.lean`. It states the results with explicit Lean types and supplies proof terms using the complete shared implementation included in the collection. The manuscript is self-contained mathematically; the Lean entry file uses those ordinary library imports rather than duplicating the full foundation.

The entry's named theorem interfaces are listed below. The namespace of each is `FinalCollection.A393856`.

- `source_existence_uniqueness`
- `source_normalized`
- `source_equation`
- `a393856_mod_five`
- `a393856_parity`

The preserved dependencies also contain all supporting constructions, coefficient identities, map-compatibility results, and inverse proofs. The collection pins Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The verification report records fresh compilation and the transitive axiom audit. Finite numerical checks corroborate the definitions and indexing; the universally quantified Lean theorems establish the all-index claims. No assertion of worldwide novelty, OEIS acceptance, publication, or completed human review follows from these checks. The prose may use elementary identities that the formal proof establishes by a different algebraic route.

References

[1] P. D. Hanna, *A393856*, The On-Line Encyclopedia of Integer Sequences. `https://oeis.org/A393856`. Source recorded in the retained proof collection, 24 October 2026.

[2] D. Goyal, *OEIS A396843 proof repository.* `https://github.com/dvgyl/oeis-a396843-proof`. Reference revision `995e55adcbe06b2a913e24f12ff73a7faf362f06`.